EXECUTIVE STAKEHOLDER SUMMARY

Real-Time Ferry Ticket Sales & Redemption Analytics for Toronto Island Park

PURPOSE OF THIS DOCUMENT

This Executive Stakeholder Summary presents high-level findings from a decade-long analysis of Toronto Island Park ferry ticket sales and redemption data. It is designed for City of Toronto government stakeholders, senior management, and policy decision-makers who require concise, actionable intelligence without deep technical detail. The full research paper, interactive Streamlit dashboard, and complete dataset are publicly available via the DOI above.

AT A GLANCE — 10 YEARS OF FERRY OPERATIONS

1. TOTAL TICKETS SOLD (2015–2025)

12,972,051

Approximately 1.3 million tickets per year — demand has grown above pre-COVID levels by 2025

2. OVERALL REDEMPTION RATE

98.6%

Near-perfect ticket utilisation — every purchased ticket is almost always used for boarding

3. NET PASSENGER MOVEMENT

+186,758

More tickets sold than redeemed over the decade — island occupancy has grown sustainably

4. PEAK DEMAND HOUR

12:00 Noon (Top window: 10:00 – 14:00)

Three hours account for ~32% of daily ticket demand — critical scheduling window

5. OFF-SEASON UTILISATION INDEX

7.4%

Off-peak hours generate only 7.4 cents of demand for every dollar generated at peak

6. SEASONAL CONCENTRATION

August = 26.4× January volume

Three summer months (Jun–Aug) account for 64.3% of annual sales — extreme seasonal asymmetry

KEY FINDINGS FOR STAKEHOLDERS

The following six findings represent the most strategically significant insights from the 10-year analysis. Each finding is grounded in verifiable data and carries direct implications for policy, operations, and investment decisions.

FINDING 1 — Demand is Extremely Concentrated in Time and Season.

The busiest three hours of the day (10:00–13:00) account for approximately 32% of all daily ticket sales. The busiest three months (June, July, August) account for 64.3% of annual volume. August alone generates 26.4 times the January volume. This compounding concentration means

that the entire annual service delivery challenge is essentially a summer-weekend-noon problem

— and it must be planned and resourced accordingly.

Implication: Ferry capacity, staffing budgets, and terminal infrastructure must be sized for summer weekend peak demand — but the same resources sit largely idle for 18 out of 24 hours and 4 out of 12 months.

FINDING 2 — A 98.6% Redemption Rate Enables Predictive Scheduling

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Nearly every ferry ticket sold results in a boarding. This near-perfect redemption rate means that real-time sales monitoring provides an accurate same-day forecast of actual passenger volume. The City does not need additional sensor infrastructure or passenger counting systems — the existing ticketing data is sufficient for operational demand forecasting.

Implication: The 1-hour rolling average of ticket sales (already computed in the dataset) provides a 45-minute advance warning of demand surges — sufficient time to pre-position additional ferry departures before queues form at the terminal.

FINDING 3 — Off-Peak Capacity is Critically Underutilised

The Off-Season Utilisation Index of 7.4% means that ferry services outside the 8:00–20:00 peak window generate only 6.9% of total annual sales. September and October record the lowest

OSUI values (4.9%) — the shoulder season sees a rapid collapse of off-peak demand even while peak volumes remain substantial. This represents either a cost reduction opportunity (rationalise off-peak service) or a revenue growth opportunity (stimulate off-peak demand).

Implication: A structured review of off-peak service frequency in September–October could generate meaningful operating cost savings without material impact on passenger access.

FINDING 4 — Net Passenger Movement Signals Return-Wave Congestion Risk

Net Passenger Movement (tickets sold minus tickets redeemed) is positive during morning hours

— indicating net outbound flow to the island. It turns sharply negative during 10:00–18:00 as pre-purchased ticket holders board en masse. Then it turns positive again at 19:00–23:00 as island visitors return. This pattern reveals a predictable return-wave congestion risk at the mainland terminal every evening between 17:00 and 20:00 on high-demand days.

Implication: Evening return-wave management — additional staff, crowd control, and ferry frequency — should be a standard operating procedure on days when 10:00 rolling average sales

exceed the 90th percentile threshold.

FINDING 5 — COVID-19 Caused a 70.7% Demand Collapse — Fully Recovered by 2023

The pandemic reduced 2020 ferry demand to 366,606 tickets — down 70.7% from 2019's 1,249,725.

By 2023, demand had recovered to 1,491,473 tickets (119% of 2019), and 2025 reached 1,507,674

— the second-highest annual total in the dataset's history. This faster-than-expected recovery confirms that underlying demand for Toronto Island access is structurally robust and that the pandemic represented a temporary suppression, not a permanent shift.

Implication: Capital planning and infrastructure investment decisions should be based on the 2023–2025 recovery trend rather than COVID-era baselines. The service is growing.

FINDING 6 — Saturdays and Sundays Drive Disproportionate System Demand

Weekends account for 40.7% of total annual sales despite representing only 28.6% of calendar days. Saturday alone generates 2.1 times the demand of Tuesday — the quietest weekday. Weekend redemption rates (102.5–105.1%) slightly exceed weekday rates, indicating a higher proportion of advance-purchased tickets on weekends. The service is fundamentally a leisure weekend service — not a commuter transit service — which has profound implications for how it

should be staffed, priced, and managed.

PERFORMANCE SNAPSHOT — KEY METRICS

KPI	Value	Benchmark	Status
Total Tickets Sold (2025)	1,507,674	+20.6% vs 2019	Above pre-COVID
Redemption Rate	98.6%	Target: ≥95%	Exceeds target
Peak Hour (12:00) Daily Share	~6.6% of daily volume	Expected: high	Concentration risk
Off-Season Util. Index	7.4%	Target: ≥15%	Below target
Weekend Sales Share	40.7%	Leisure benchmark	Healthy leisure use
August vs January Ratio	26.4 : 1	Nordic avg: 10:1	High seasonality
Post-COVID Recovery	119% of 2019 by 2023	Target: 100%	Fully recovered
Anomaly Rate (|Z|>3)	2.6% of intervals	Alert threshold	Monitor proactively

STRATEGIC RECOMMENDATIONS

Based on the 10-year data analysis, eight strategic recommendations are presented below.

They are prioritised by implementation ease and potential impact.

IMMEDIATE ACTIONS (0–3 months)

01 | Deploy the Analytics Dashboard Organisation-Wide

The Streamlit dashboard (DOI: 10.5281/zenodo.19132415) provides role-specific views for Operations Teams, Policy Planners, and Management Stakeholders. Deployment to Streamlit Cloud takes approximately 2–4 hours and provides browser-accessible analytics to all staff without any local software installation. Cost: Near-zero.

02 | Implement Rolling Average Demand Alert Protocol

Configure a threshold alert (1-hour rolling average sales > 150 tickets/interval = 600/hr) to notify operations staff before terminal queues form. The 1-hour rolling average is pre-computed in the dataset. This requires only a monitoring screen and an SOP document. Cost: Minimal.

SHORT-TERM ACTIONS (3–12 months)

03 | Review Off-Peak Service Frequency in September–October

The September–October OSUI of 4.9% — the lowest in the dataset — indicates systematic

idle capacity during the fall shoulder season. A reduction of one return trip during 21:00–07:00 on weeknights in these months could generate meaningful cost savings. Requires route-level data disaggregation to confirm Ward's Island service is unaffected.

04 | Launch Off-Season Demand Stimulation Program

January–March generates only 130,000–200,000 tickets per month. Targeted promotional programs — winter photography tours, school group visits, senior discount rates — could increase winter volumes by 20–40%. Target: Increase January OSUI from 10.1% to 15% within 3 years.

05 | Pilot Time-of-Day Pricing for Summer Peak Windows

A modest premium (+15%) for 10:00–14:00 tickets on summer weekends, offset by a discount (-10%) for 08:00–10:00 and 15:00–18:00 bookings, could smooth the demand curve. Reduces terminal congestion at the core peak while maintaining total visitor volumes.
Pilot in Summer 2026 with pre/post volume analysis.

LONG-TERM INVESTMENTS (12–36 months

06 | Add Route-Level Identifiers to the Ticketing System

Currently Centre Island, Hanlan's Point, and Ward's Island are aggregated into a single count. Adding route identifiers enables route-specific scheduling, residential demand analysis, and targeted marketing. Required at next ticketing system upgrade.

07 | Integrate Weather Data for Demand Forecasting

Demand is approximately 40% higher on 25°C days versus 15°C days. Integrating Environment Canada open weather data (free API) would improve same-day scheduling accuracy by an

estimated 15–25%.

08 | Connect Real-Time Data Feed & Build ML Forecasting Model

Phase the dashboard from historical analytics to live operational monitoring. Build an XGBoost/LightGBM model for 24–72 hour ahead demand forecasts for advance scheduling.

EXPECTED IMPACT MATRIX

Recommendation	Efficiency	Cost Impact	Passenger Exp.	Priority
01 Deploy Dashboard	High	Negligible	Indirect	Now
02 Rolling Avg Alerts	High	Negligible	Reduces queuing	Now
03 Off-Peak Review	Medium	Cost saving	Minimal impact	3 months
04 Off-Season Stimulation	Medium	Revenue gain	Improves access	6 months
05 Time-of-Day Pricing	High	Revenue neutral	Reduces crowds	12 months
06 Route-Level Data	Very High	Low cost	Route-specific	Next upgrade
07 Weather Integration	High	Low cost	Fewer surprises	18 months
08 Real-Time ML	Very High	Medium invest.	Improved reliab.	24 months

DASHBOARD & ANALYTICS PLATFORM

- The interactive Streamlit dashboard serves three distinct user groups:

- Operations Team: Hourly demand bars, Z-score anomaly detection, rolling averages, Day×Hour heatmaps. Use: Monitor a summer Saturday in real-time.

- Policy Planners: Year-over-year trends, seasonal comparison, Off-Season Utilisation Index charts, Year×Month heatmaps. Use: Compare COVID vs recovery.

- Management Stakeholders: High-level KPI cards, annual bar charts, monthly trend lines. Use: Board reporting and annual performance review.

- ACCESS: https://doi.org/10.5281/zenodo.19132415
- Includes: Research Paper | Executive Summary | dashboard.py | Cleaned_dataset.csv

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CONCLUSION

This project demonstrates that the Toronto Island Park ferry ticketing data — already collected and publicly available through the City of Toronto Open Data Portal — contains substantial untapped operational intelligence. The ten-year analysis, interactive dashboard, and eight strategic recommendations collectively represent an opportunity to:

- Reduce terminal congestion through rolling-average-triggered dynamic scheduling
- Generate cost savings by rationalising off-peak service frequency in the fall shoulder season
- Increase revenue through off-season demand stimulation targeting the 7.4% utilisation gap
- Improve passenger experience through time-of-day pricing that smooths the noon-hour spike
- Support data-driven capital investment based on the confirmed post-COVID recovery trajectory

Most importantly, this analysis was completed using entirely open-source tools (Python, pandas,

Streamlit, Plotly) and open government data — at minimal cost. The same analytical framework can be replicated across other City of Toronto transit services with comparable data streams.

"Data-driven decisions improve ferry scheduling, reduce terminal congestion, and deliver a better experience for the millions of Torontonians who use this beloved public service each year."